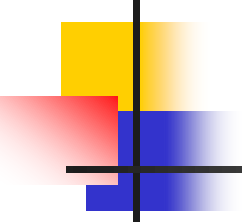


Lecture 2

Data Communications



Introduction to Data Communications

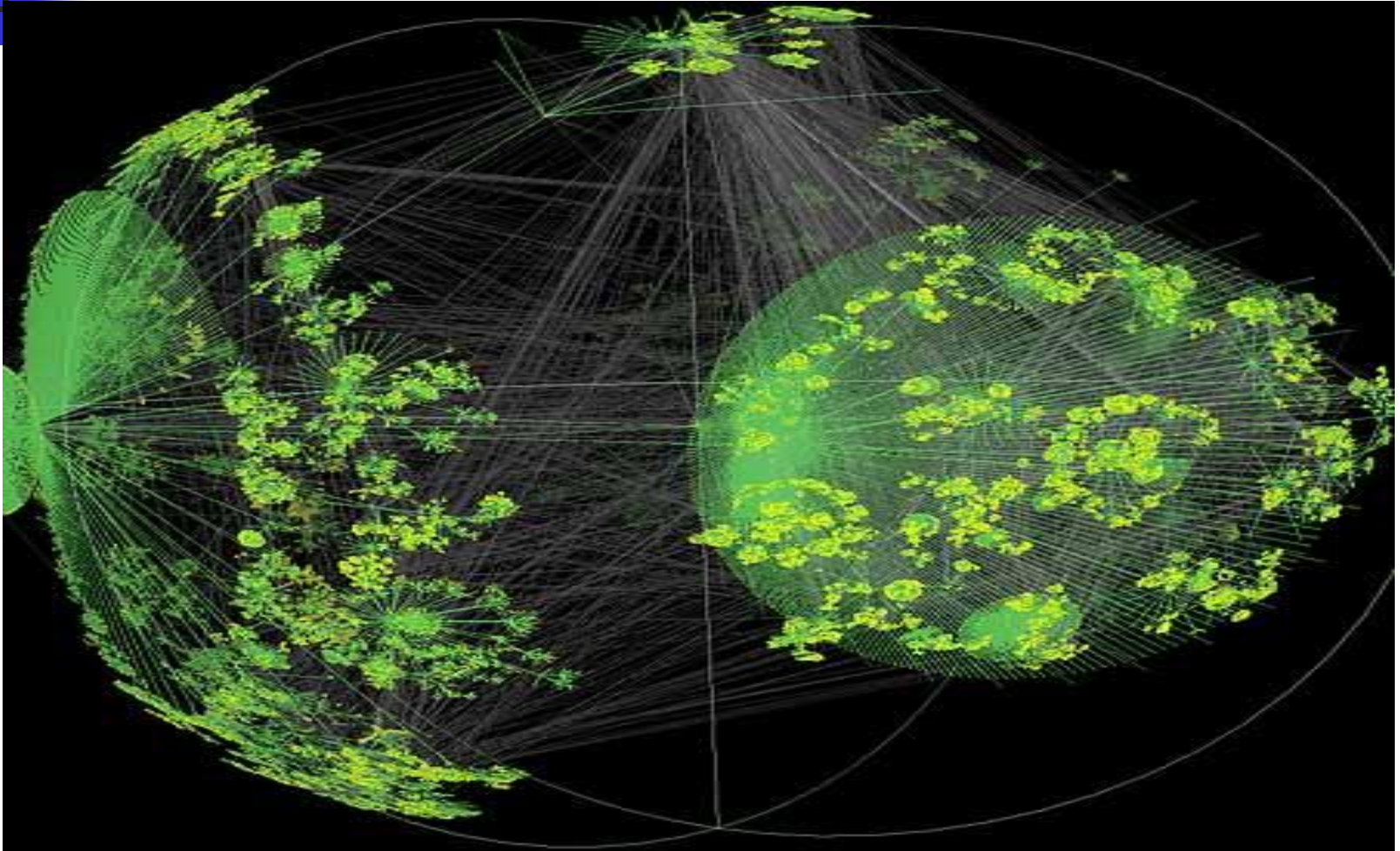
- ❑ The word **data** refers to information presented in whatever form is agreed upon by the parties creating and using the data.
- ❑ **Communication system** made up of a combination of hardware and software
- ❑ **Data communications** are the exchange of data between two devices via some form of transmission medium such as a wire cable.



Cont.

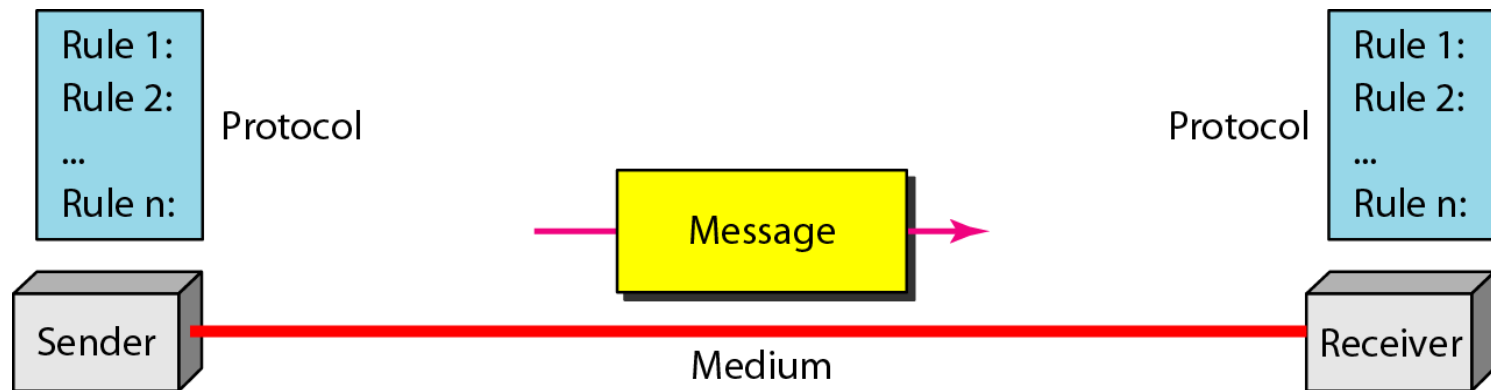
- Effectiveness of data communication system depends on:
 1. **Delivery** : The system must deliver data to correct destination. Data received by the indented user only
 2. **Accuracy**: The system must deliver data accurately (no change).
 - Data changed & uncorrected is unusable
 3. **Timeliness**: The system must deliver data in timely manner
 - Data arrived late are useless
 - In the same order (video and audio) & without delay (Real time transmission)
 4. **Jitter**: Variation in the packet arrival time (uneven quality in the video is the result)

Cont.



Components

- A data communication system is made up of **five** components



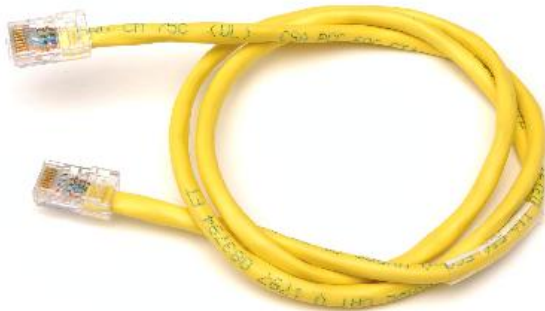


Components

1. **Message**: the information (data) to be communicated
 - Consist of text, numbers, pictures, audio, or video
2. **Sender**: the device that sends the data message
 - Computer, workstation, telephone handset, video camera, ...
3. **Receiver**: the device that receives the message
 - Computer, workstation, telephone handset, television,

Components

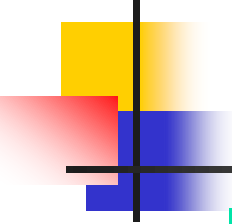
4. **Medium:** The physical path by which a message travels from sender to receiver
- twisted pair, coaxial cable, fiber-optic, radio waves





Components

5. **Protocol:** a set of rules that govern data communications
- An agreement between the communicating devices
 - Devices may be connected but not communicating (no protocol)
 - Arabic speaker with Japanese speaker



Data Representation

Text

Numbers

Images

Audio

Video



Data Representation

■ Text:

- Sequence of bits (0s or 1s)
- Different sets of patterns to represent text symbols (each set is called: *code*)
- ASCII: 7 bits (128 symbols) **Unicode**
- common coding system today is:
- Unicode uses: **32 bits** to represent a symbol or character in any language (4,294,967,296)

10101010 10101010 10101010 10101010 10101010 10101010 10101010



Data Representation

- **Numbers:**
 - Represented by bit patterns
 - The number is directly converted to a **binary** number



Data Representation

- **Images:**

- Represented by bit patterns
- A matrix of **pixels**
- Resolution: size of the pixels
- High resolution: more memory is needed
- Each pixel is assigned a bit pattern
 - 1-bit pattern (black and white dots image)
 - 2-bit pattern (4 levels of gray)
 - **RGB** (color images)



Data Representation

- **Audio:**

- Continuous not discrete
- Change to digital signal

- **Video:**

- Recording or broadcasting of a picture or movie
- Change to digital signal

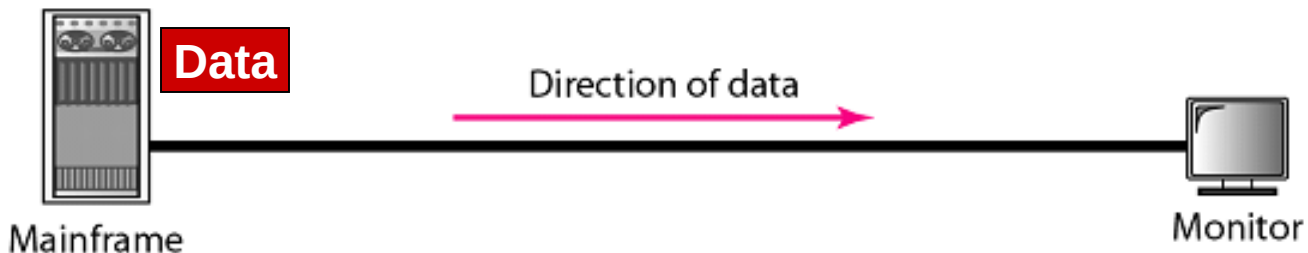


Data Flow

- Communication between two devices can be:
 - Simplex
 - Half-Duplex
 - Full-Duplex

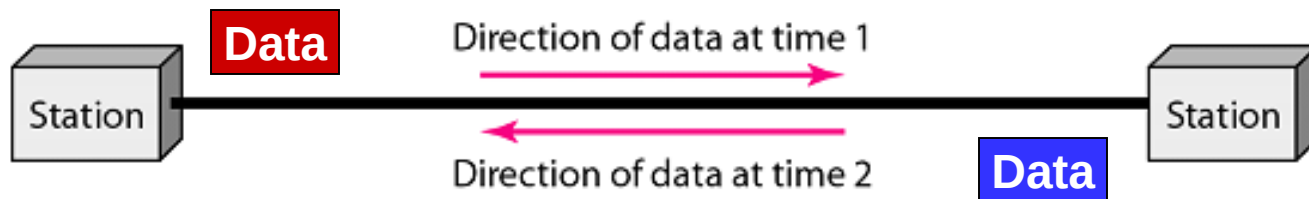
Data Flow

- **Simplex** (one way street)
 - The communication is unidirectional
 - Only one device on a link can transmit; the other can only receive
 - Use the entire capacity of the channel to send data
 - Example: Keyboards, Monitors



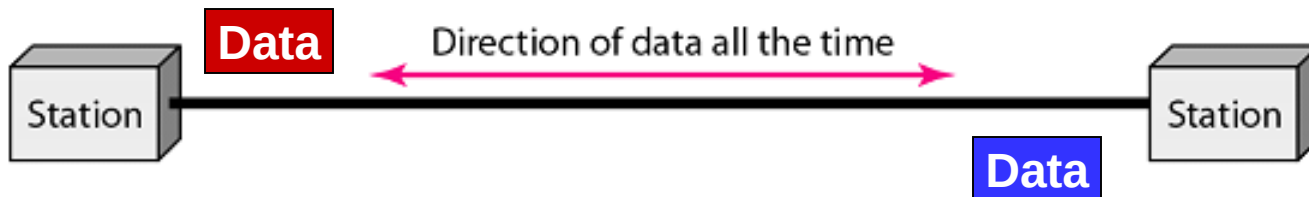
Data Flow

- **Half-Duplex** (one-lane with two-directional traffic)
 - Each station can both transmit and receive, but not at the same time
 - When one device is sending, the other can only receive, and vice versa
 - The entire capacity of a channel is taken over by the transmitting device
 - Example: Walkie-talkies



Data Flow

- **Full-Duplex (Duplex)** (two-way street)
 - Both stations can transmit and receive at same time
 - Signals going in either direction sharing the capacity of the link
 - Sharing can occur in two ways:
 - Link has two physically separate transmission paths
 - One for sending and the other for receiving
 - The capacity of the channel is divided between signals





Networks

- **Network** : A set of devices (**nodes**) connected by communication links
 - Node** : computer, printer, ...
- **Distributed Processing** :
 - Most networks used it
 - Task is divided among **multiple computers** instead of one single large computer



Networks

■ Network Criteria

- Network must meet a certain number of criteria
- The most important of the network criterions are:
 - Performance
 - Reliability
 - Security



Networks

- Performance

- **Transit time:** A mount of time required for a message to travel from one device to another
- **Response time:** Elapsed time between an inquiry and a response



Networks

■ Performance

■ Performance depends on :

- 1- **Number of users**: large number slow response time.
- 2- **Type of transmission medium**: fiber-optic cabling faster than others cables.
- 3- **Capabilities of the connected hardware**: affect both the speed and capacity of transmission.
- 4- **Efficiency of the software**: process data at the sender and receiver and intermediate affects network performance.



Networks

- Performance

- Performance is evaluated by two contradictory networking metrics:
 - **Throughput** (high): a measure of how fast we can actually send data through a network
 - **Delay** (low)



Networks

■ Reliability

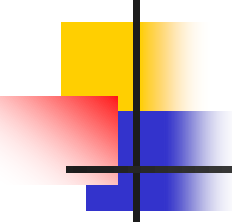
- Reliability is measured by:
 1. Frequency of failure
 2. Recovery time of a network after a failure
 3. Network's robustness in a catastrophe:
protect by good back up network system



Networks

- Security

- Protecting data from unauthorized access
- Protecting data from damage and development
- Implementing policies and procedures for recovery from breaches and data losses (Recovery plan)



Networks

- Physical Structures:

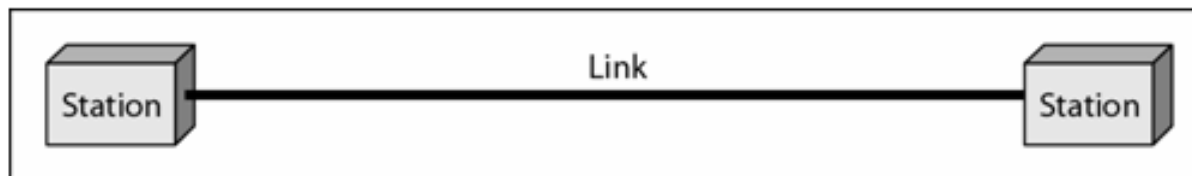
- Type of connection

- **Network**: Two or more devices connected through links
 - **Link**: Communication pathway that transfers data from one device to another
 - Two devices must be connected in some way to the same link at the same time. Two possible types:
 - Point-to-Point
 - Multipoint

Networks

- Point-to-Point

- Dedicated link between two devices
- Entire capacity of the link is reserved for transmission between those two devices
- Use an actual length of wire or cable



a. Point-to-point

Networks

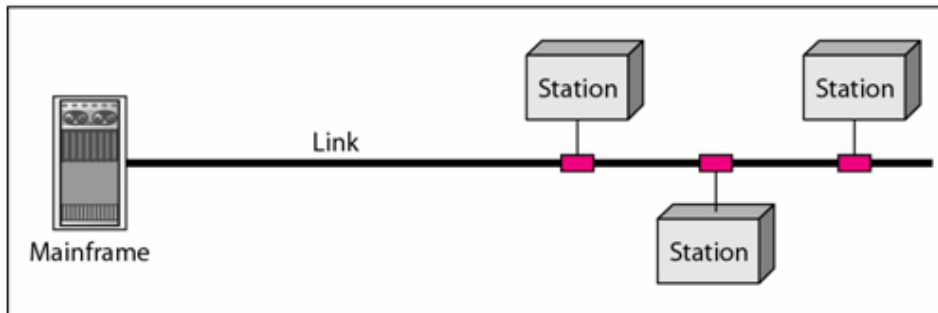
- Point-to-Point
 - Other options, such as microwave or satellite is possible
 - Example: Television remote control



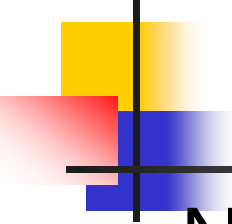
Networks

Multipoint (multidrop)

- More than two devices share a single link
- Capacity is shared
- Channel is shared either spatially or temporally
 - Spatially shared: if devices use link at same time
 - Timeshare: if users must take turns



b. Multipoint



Networking Devices(Nodes)

1. NIC Card
2. Repeater
3. Hub
4. Switch
5. Bridge
6. Router
7. Gateway
8. Firewall

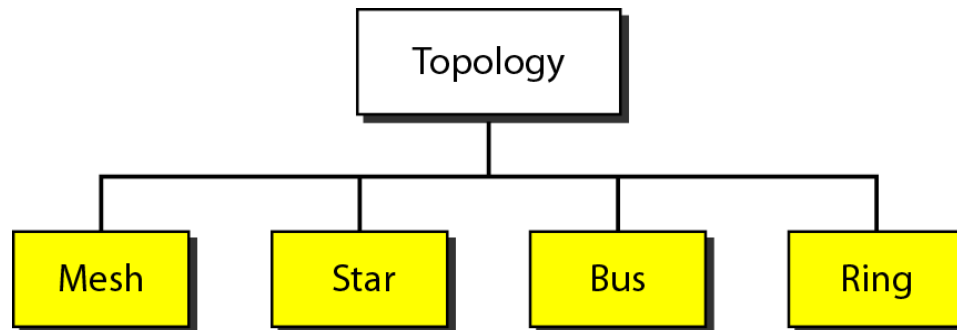


Network Topologies

- ❑ The topology of a network is the geometric representation of the relationship of all the links and linking devices (nodes) to one another.
- ❑ Network topology defines the structure of the network.
- ❑ A goal of any topology
 1. high **throughput** (bandwidth)
 2. low **latency**
- A. **Physical topology**:- It define the actual layout of the wire or media.
Four topologies : **Mesh**, **Star**, **Bus**, and **Ring**
- B. **Logical topology**:- It defines how the hosts access the media to send data.
 1. Broadcast
 2. Token passing
- C. **Hybrid Topology**



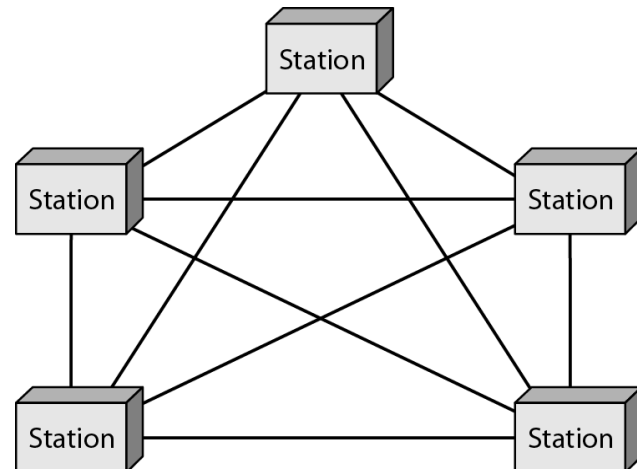
Physical Topology



Physical Topology

- Mesh

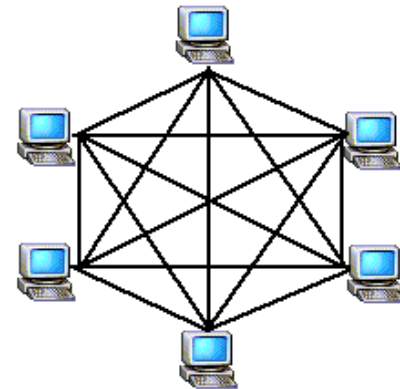
- Every link is dedicated point-to-point link
- The term dedicated means that the link carries traffic only between the two devices it connects



Physical Topology

- Mesh

- To link n devices fully connected mesh has: $n(n - 1) / 2$ physical channels (Full-Duplex)
- Every Device on the network must have $n - 1$ ports



Physical Topology

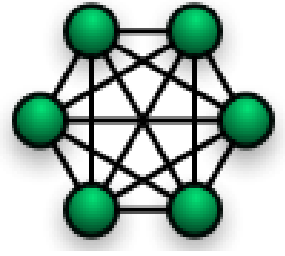
- Mesh

- Example:

8 devices in mesh has links: $n(n-1) / 2$

number of links = $8 (8-1)/2 = \mathbf{28}$

number of ports per device = $n - 1 = 8 - 1 = \mathbf{7}$

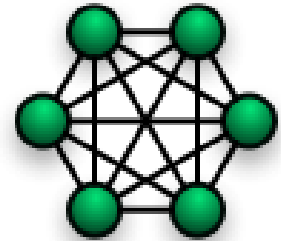


Physical Topology

- Mesh

- Advantages

- Each connection carry its own data load (no traffic problems)
 - A mesh topology is robust
 - Privacy or security
 - Fault identification and fault isolation



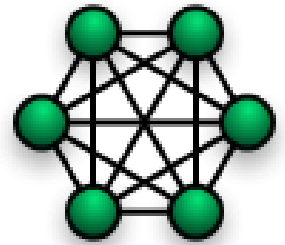
Physical Topology

- Mesh:

- Disadvantages

- Big amount of cabling
 - Big number of I/O ports
 - Installation and reconnection are difficult
 - Sheer bulk of the wiring can be greater than the available space
 - Hardware connect to each I/O could be expensive

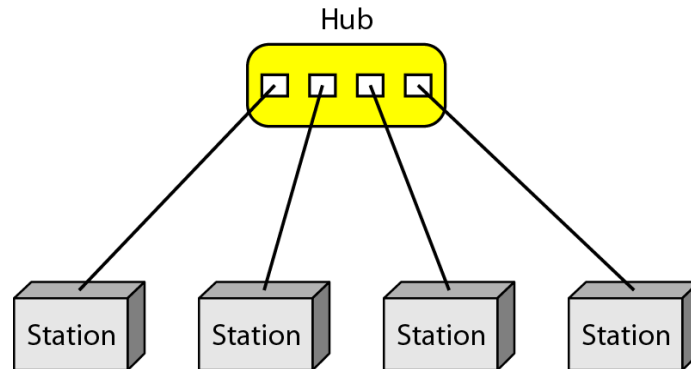
- Mesh topology is implemented in a limited fashion; e.g., as backbone of hybrid network



Physical Topology

- Star:

- Dedicated point-to-point to a central controller (Hub)
- No direct traffic between devices
- The control acts as an exchange



Physical Topology

■ Star

■ Advantages

- Less expensive than mesh
(1 Link + 1 port per device)
- Easy to install and reconfigure
- Less cabling
- Additions, moves, and deletions required one connection
- Robustness : one fail does not affect others
- Easy fault identification and fault isolation



Physical Topology

- Star

- Disadvantages

- Dependency of the whole topology on one single point (hub)
 - More cabling than other topologies (ring or bus)

- Used in LAN



Physical Topology

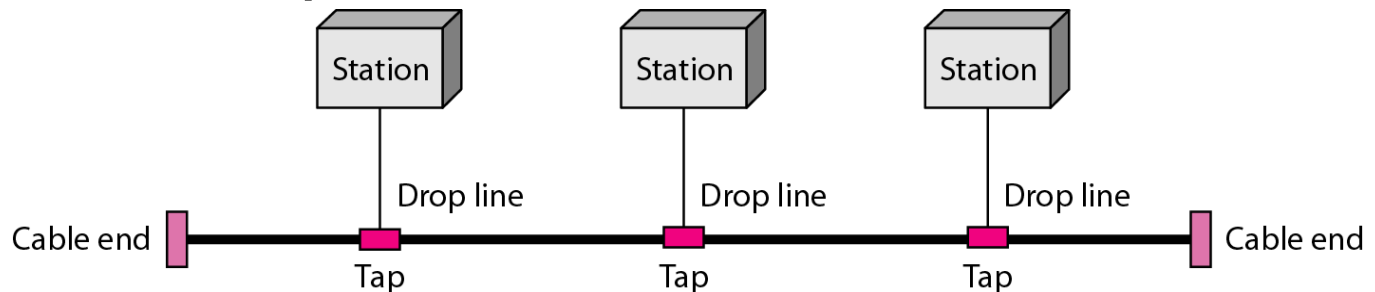
- Bus
 - It is multipoint
 - One long cable acts as a backbone
 - Used in the design of early LANS, and Ethernet LANs



Physical Topology

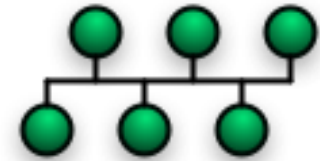
Bus

- Nodes connect to cable by drop lines and taps
- Signal travels along the backbone and some of its energy is transformed to heat
- Limit of number of taps and the distance between taps



Physical Topology

■ Bus



■ Advantages

- Ease of installation
- Less cables than mesh, star topologies

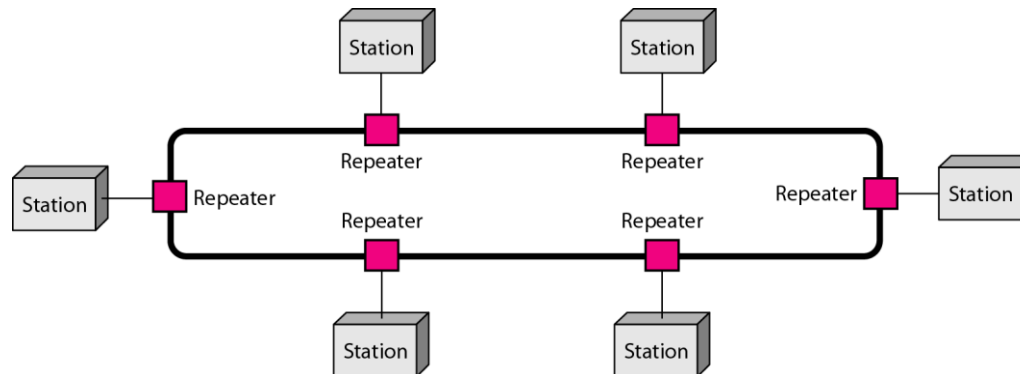
■ Disadvantages

- Difficult reconnection and fault isolation (limit of taps)
- Adding new device requires modification of backbone
- Fault or break stops all transmission
- The damaged area reflects signals back in the direction of the origin, creating noise in both directions

Physical Topology

■ Ring

- Each device has dedicated point-to-point connection with only the two devices on either side of it
- A signal is passed along the ring in one direction from device to device until it reaches its destination
- Each devices incorporates a Repeater



Physical Topology

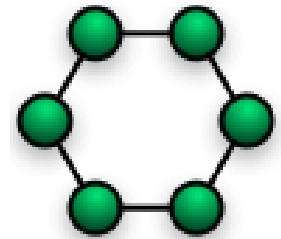
- Ring

- Advantages

- Easy of install and reconfigure
 - Connect to immediate neighbors
 - Move two connections for any moving (Add/Delete)
 - Easy of fault isolation

- Disadvantage

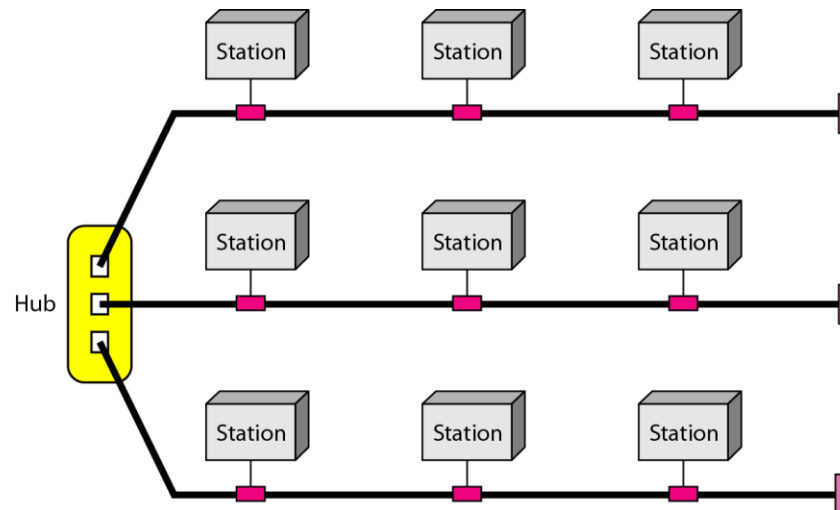
- Unidirectional
 - One broken device can disable the entire network. This weakness can be solved by using a dual ring or a switch capable of closing off the break



Physical Topology

■ Hybrid Topology

- Example: having a main star topology with each branch connecting several stations in a bus topology





Categories of Networks

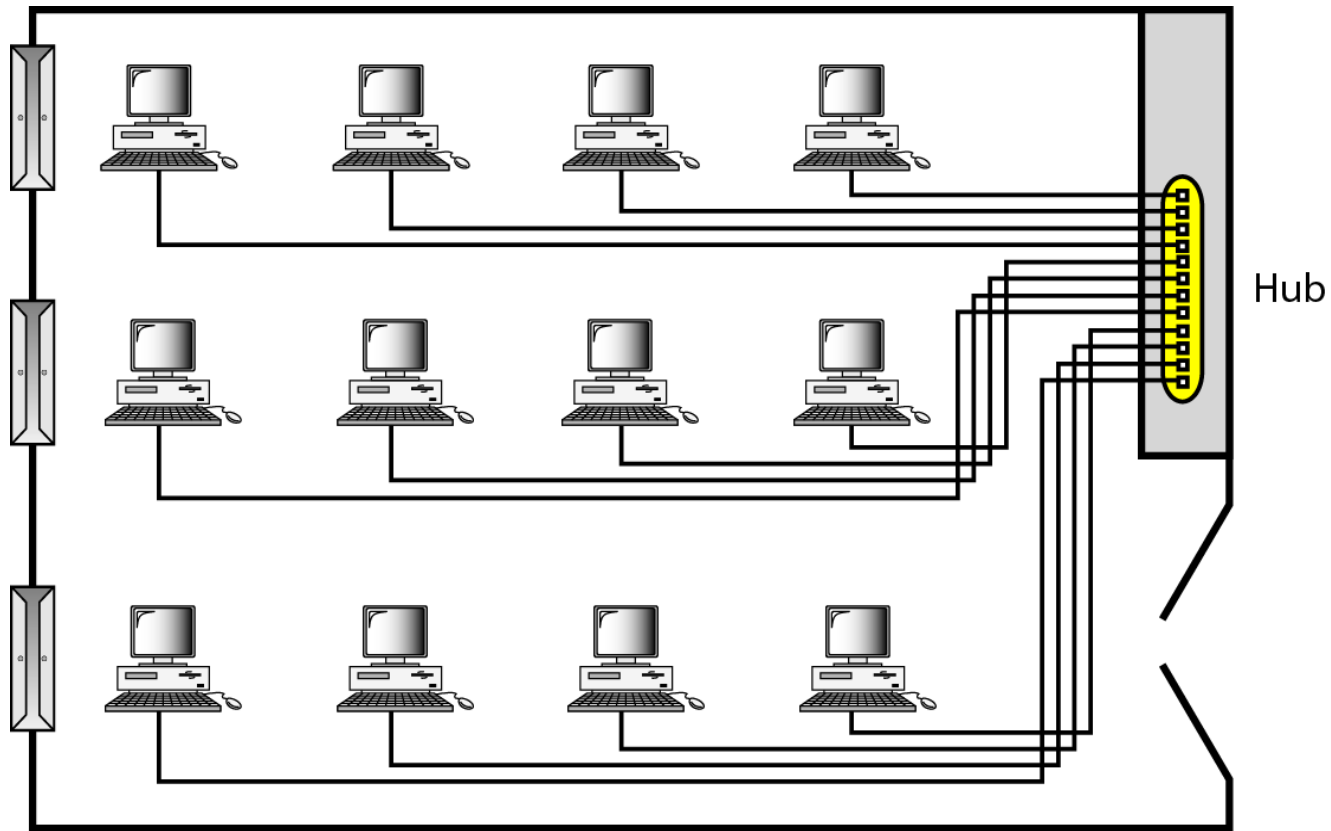
- Network Category depends on its size
- Two primary categories
 - **LAN**: Covers area < 2miles
 - **WAN**: Can be worldwide
 - **MAN**: Between LAN & WAN, span 10s of miles



Local Area Network (LAN)

- Privately owned
- Links devices in the same office, building, or campus
- Simple LAN: 2 PCs & 1 printer in home or office
- Size is limited to a few kilometers
- Allow resources to be shared (hardware, software, or data)

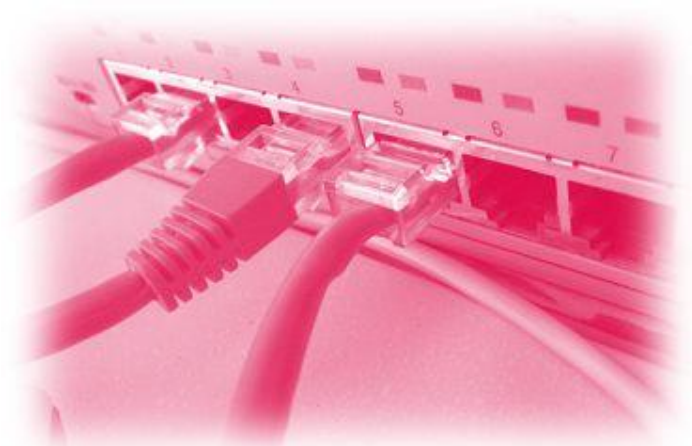
Local Area Network (LAN)



An isolated LAN connecting 12 computers to a hub in a closet

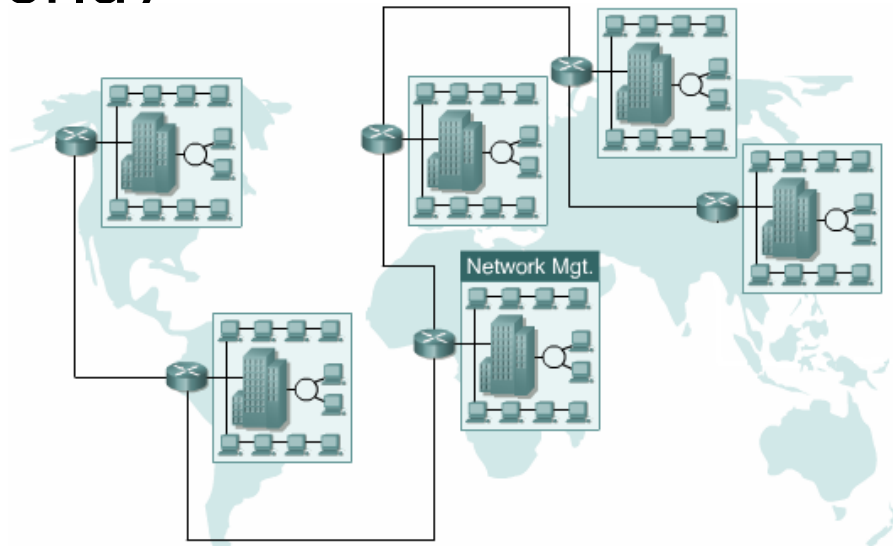
Local Area Network (LAN)

- LAN is distinguished by:
 - Size (# users of OS, or licensing restrictions)
 - Transmission medium (only one type)
 - Topology (bus, ring, star)
- Data Rates (speed):
 - Early: 4 to 16 Mbps
 - Today: 100 to 1000 Mbps



Wide Area Networks (WAN)

- Provides long-distance transmission of data over large geographic areas (country, continent, world)

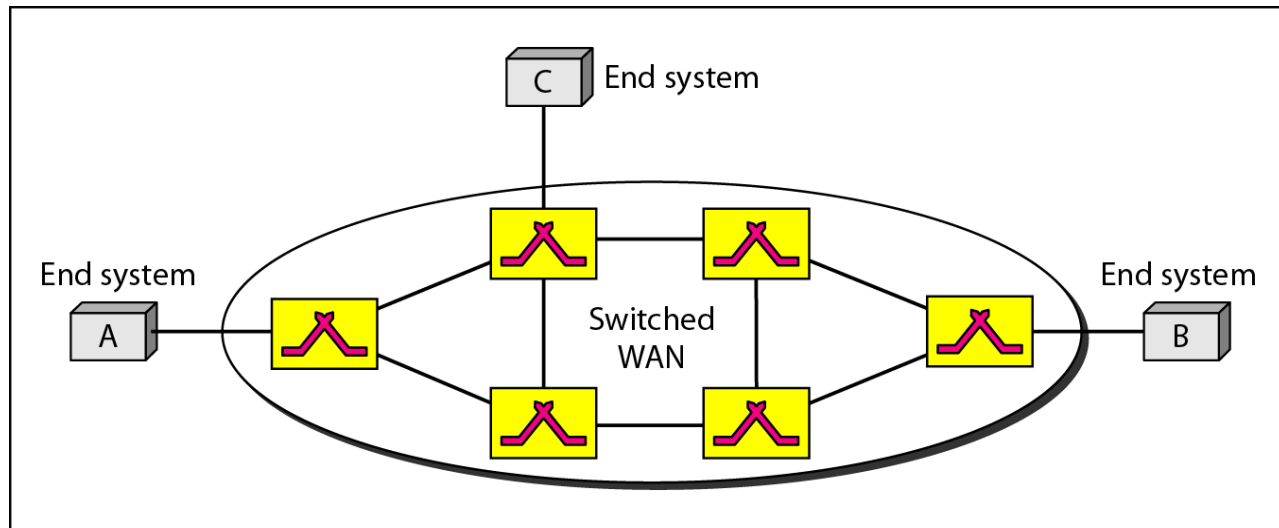




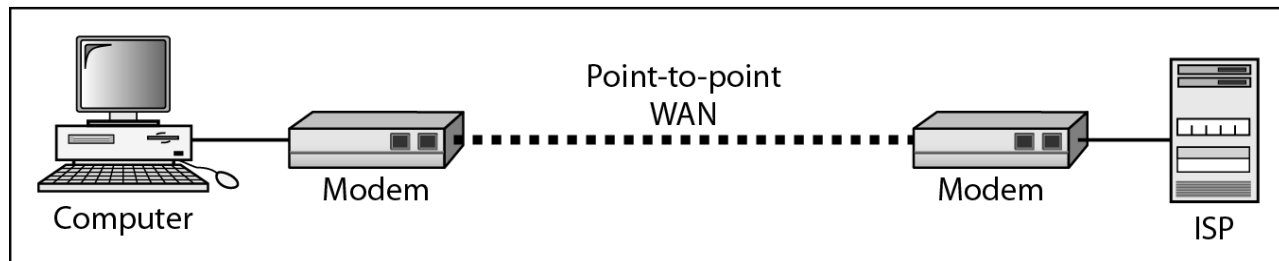
Wide Area Networks (WAN)

- Switched WAN
 - Backbone of the Internet
- Dialup line point-to-point WAN
 - Leased line from a telephone company

Wide Area Networks (WAN)



a. Switched WAN



b. Point-to-point WAN

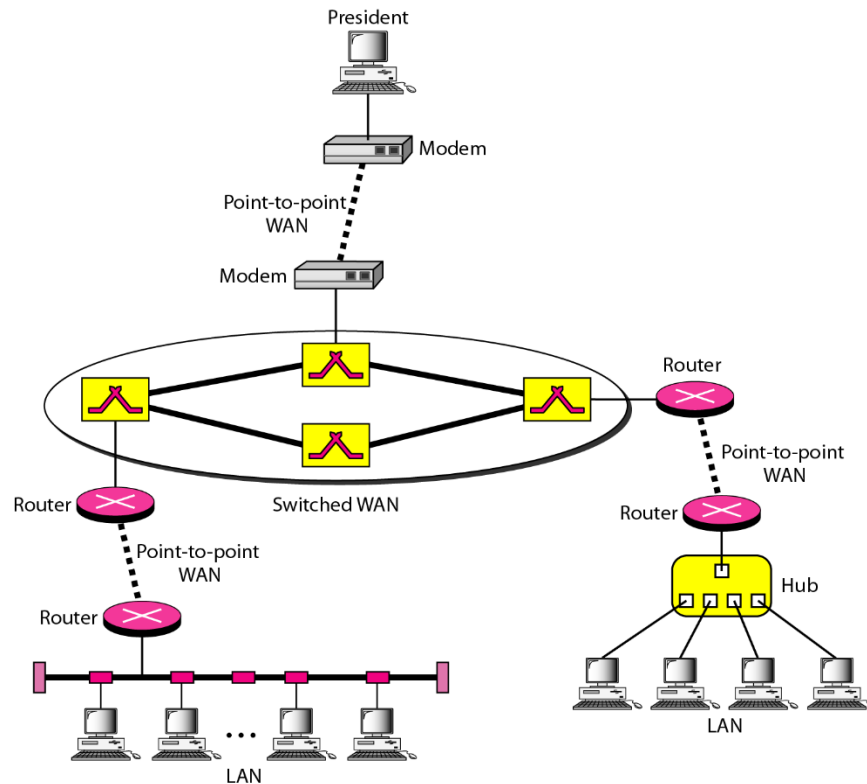


Metropolitan Area Networks (MAN)

- Size between LAN and WAN
- Inside a town or a city
- Example: the part of the telephone company network that can provide a high-speed DSL to the customer

Interconnection of Networks: Internetworks

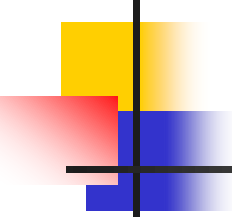
- Two or more networks connected together



The Internet

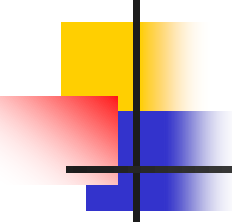
- **Internet** has revolutionized many aspects of our daily lives.
- It has affected the way we do business as well as the way we spend our leisure time.
- Internet is a communication system that has brought a wealth of information to our fingertips and organized it for our use
- An **internet** is 2 or more networks that can communicate with each other
- The **Internet** is a collaboration of more than hundreds of thousands of interconnected networks





The Internet

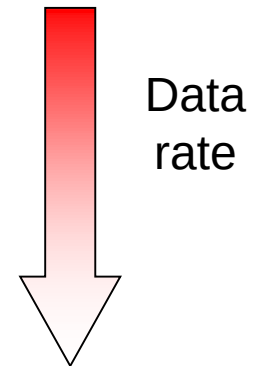
- An internet (small i) is two or more networks
- Notable internet is called the Internet (hundreds of thousands interconnected networks)
 - Private individuals + government agencies + school + research facilities + Corporations + libraries in more than 100 countries
- This communication system came in 1969
- Mid-1960 (ARPA) Advanced Research Projects Agency in (DOD) was interested to connect mainframes in research organizations
- 1967, ARPA presented its ideas for ARPANET
 - Host computer connecting to (IMP) interface message processor.
 - Each IMP communicate with other IMP
- 1969, four nodes (universities) connected via IMPs to form a network
 - Software (NCP) Network Control Protocol provided communication between the hosts.
- 1972, Vint Cerf and Bob Kahn invented (TCP) Transmission Control Protocol
- Later TCP was split to (TCP) Transmission Control Protocol and (IP) Internetworking Protocol



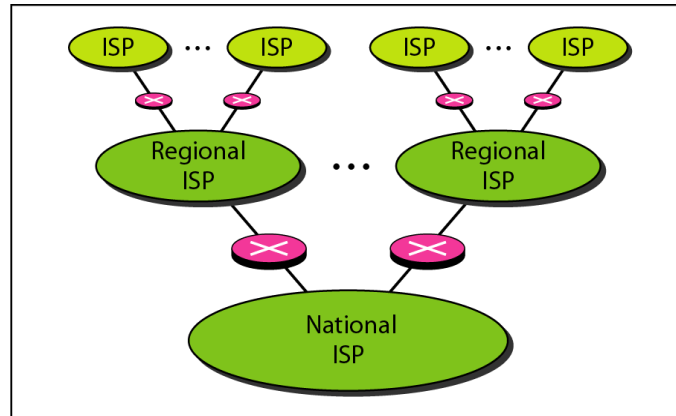
The Internet

■ Internet Today

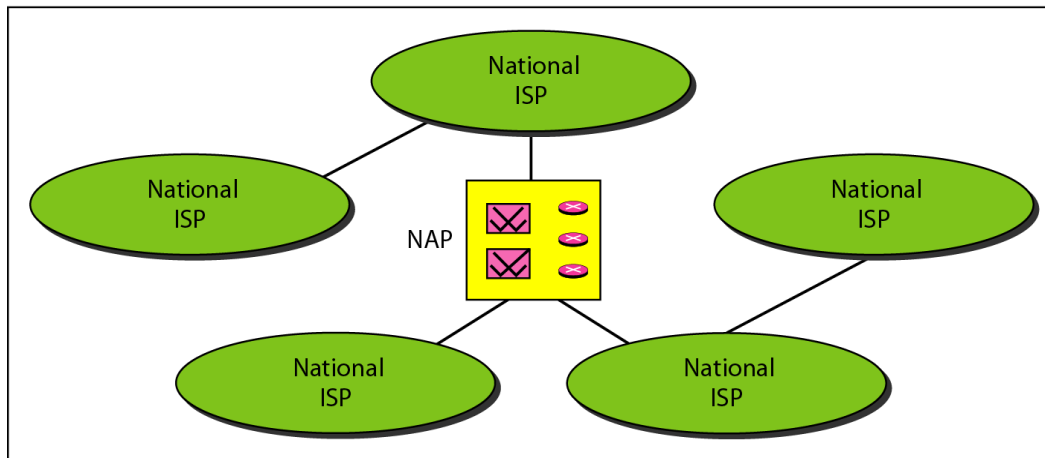
- Made of many LANs and WANs
- Every day new networks area added and removed
- Internet services Providers (ISPs) offer services to the end users
 - International service providers
 - National service providers
 - Regional service providers
 - Local service providers



The Internet



a. Structure of a national ISP



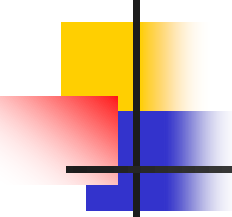
b. Interconnection of national ISPs

Hierarchical organization of the Internet



Protocols and Standards

- Protocol synonymous with rule
- Standards: agreed-upon rules
- Protocols
 - A protocol is a set of rules that govern data communications
 - Defines What, How, and When it is communicated



Protocols and Standards

- Elements of a protocol:
 - **Syntax:** structure or format of data
 - Example: 8-bits address of sender, 8-bits address of receiver
 - **Semantics:** meaning of each section of bits
 - Example: Does the address is a route to be taken or the final destination of the message
 - **Timing:** when data should be sent and how fast they can be sent
 - Example: sender produces data at 100 Mbps but the receiver can process data at only 1 Mbps ⇒ overload and data loose



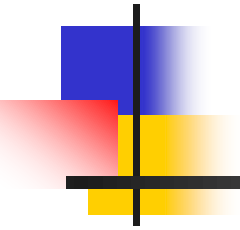
Standards

- Essential in creating and maintaining an open and competitive market for equipment manufactures
- Guaranteeing national and international interoperability of data and telecommunication technology and processes
- Providing guidelines to manufacturers, vendors, government agencies, and other service providers to ensure the kind of interconnectivity necessary in today's marketplace and in international communications



Standards

- Standards are developed through the cooperation of:
 - Standards Creation Committees
 - ISO, ITU-T, CCITT, ANSI, IEEE, EIA
 - Forums
 - Created by special-interest groups
 - Present their conclusions to the standards bodies
 - Regulatory Agencies
 - Ministry of Information, Communication and Information Technology (MICIT)
 - Purpose: Protecting the public by regulating radio, television, and communication



Thank You!